

SACHIN KALYANI

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OBJECTIVE

PhD student in Computational Biology with a background in biochemistry and molecular biology. I specialize in applying deep learning and statistical methods to genomic and behavioral data, with a focus on neurodevelopmental disorders. My objective is to develop interpretable AI models for early diagnosis and therapeutic understanding.

EDUCATION

- **Institute of Bioinformatics and Applied Biotechnology (IBAB), Bengaluru**
Ph.D. in Computational Biology Feb 2024 – Present
– Research focus: Studying genomics of social interaction in autism spectrum disorder employing machine and deep learning approaches
- **Pondicherry University**
M.Sc. Biochemistry and Molecular Biology 2020 – 2022
– CGPA: 8.9/10
- **Aurora's Degree and PG College, Hyderabad**
B.Sc. Biochemistry, Microbiology, Chemistry 2017 – 2020
– CGPA: 8.9/10
- **Sri Chaitanya Junior Kalasala, Hyderabad**
Higher Secondary Education (HSC) 2015 - 2017
– Score: 95.6%
- **Brilliant Grammar High School**
Secondary School Certificate (SSC) 2014 - 2015
– CGPA: 8.9/10

GRADUATE COURSEWORK

Mathematics for Biologists, Machine Learning, Deep Learning and Applications, Research Methodology, Research and Publication Ethics

EXPERIENCE

- **Syngene IBAB Certificate Program in Artificial Intelligence in Life Sciences (SAILS)**
Teaching Assistant – Python, Machine Learning, Deep Learning April 2026 – July 2026
– Assisted lab sessions, and supported student projects in Python, scikit-learn, and PyTorch.
- **Institute of Bioinformatics and Applied Biotechnology (IBAB), Bengaluru**
Teaching Assistant – Machine Learning Jan 2026 – May 2026
– Assisted lab sessions, and supported student projects in Python, scikit-learn.
- **Institute of Bioinformatics and Applied Biotechnology (IBAB), Bengaluru**
Teaching Assistant – Deep Learning July 2025 - November 202
– Assisted lab sessions, taught GPU usage, prepared tutorials, and supported student projects in Python and PyTorch.
- **Institute of Bioinformatics and Applied Biotechnology (IBAB), Bengaluru**
Teaching Assistant – Machine Learning Jan 2025 – May 2025
– Assisted lab sessions, prepared tutorials, and supported student projects in Python and scikit-learn.

SKILLS

- **Programming Languages:** Python, R, SQL, Shell scripting
- **Machine Learning / Deep Learning:** scikit-learn, SHAP, PyTorch
- **Data Science Tools:** NumPy, Pandas, Matplotlib, Seaborn
- **Molecular Biology Techniques:** Western Blotting, Gel Electrophoresis, MTT Assay, DNA/RNA Isolation
- **Research:** Differential Expression, Network Analysis, miRNA-mRNA Prediction, Representation Learning

ACHIEVEMENTS

- First Prize – Scientific Writing, Carl Neuberg celebrations, Aurora’s Degree and PG College
- Best Poster Award – ICEBG-2023 (miRNA-mRNA interactions in Parkinson’s)
- Qualified – DBT-BET Category II, GATE-XL and GATE-BT, 2023

PROJECTS

- **Predictive Utility of Vineland-3 Adaptive Behavior Scales for Cognitive Function, Social Communication, and Motor Skills Deficit**

Poster at Big Data Biology Symposium – IISc, Bangalore

May 2025

- Modeled adaptive behavior subdomains (social, cognitive, motor) using statistical and machine learning approaches.
- Presented poster at national-level symposium on AI and genomics.

- **Construction and Bioinformatic Analysis of miRNA-mRNA Regulatory Network in Alzheimer’s Disease; Evaluating the Effect of Thymoquinone on H₂O₂-induced Stress in HEK-293T Cell Line**

Academic Thesis – Pondicherry University

Jan 2022 – Jun 2022

- Studied neuroprotective effect of thymoquinone on H₂O₂-induced oxidative stress.
- Constructed and analyzed miRNA-mRNA interaction networks to identify dysregulated pathways.

PUBLICATIONS

- Delineating the interactions among mutual miRNAs and target genes associated with Parkinson’s disease, endoplasmic reticulum stress and autophagy pathways: a computational analysis
doi:<https://doi.org/10.1093/intbio/zyag005>

WORKSHOPS & TRAINING

- **Genomic Large Language Models Workshop**

Pre-Symposium at Big Data Biology Symposium, IISc

May 2025

- Hands-on experience with genomic LLMs and transformer architectures.
- Learned applications in variant interpretation and protein modeling.

LANGUAGES

English: Professional working proficiency

Hindi: Professional working proficiency

Telugu: Professional working proficiency

Marathi: Native / bilingual proficiency